



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 03

- Title:** Implement control structures and form validation; create a grading system based on user-entered marks.
- Software/Tools Required:** HTML, JavaScript, CSS, VS Code.
- Theory:** This program implements a **Student Grading System** using **HTML, CSS, and JavaScript**. It accepts the student's name and marks for five subjects through a form. JavaScript performs **form validation** to ensure that all fields are filled and that the marks entered are within the valid range (0–100). The program then calculates the **total marks, percentage, grade, and pass/fail status** using **control structures** (if, else if, else) and **loops** (for). The result is displayed dynamically on the webpage without reloading it, providing an interactive user experience.

4. Output:

Grading System

file:///C:/Users/Hp/Javascript/1189Practical03.html

Rutva

Student Grading System

76

Submit

Marks: 76
Grade: B



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CASE STUDY 03

Case Study Title: Scholarship Eligibility Checker Using JavaScript

Code:

```
Welcome Practical03.html pract04.html department.html billing.html index.html
grades.html > html
2 <html lang="en">
3 <head>
8 <style>
66 #result{
67   margin-top:20px;
68   padding:15px;
69   background: #f4f4f4;
70   border-radius:8px;
71   text-align:center;
72   font-size:18px;
73 }
74 </style>
75
76 </head>
77 <body>
78
79 <div class="container">
80
81 <h2>Scholarship Eligibility</h2>
82
83 <label>Enter Marks (0-100)</label>
84
85 <input type="number" id="marks" placeholder="Enter Marks">
86
87 <button onclick="checkScholarship()">Check Eligibility</button>
88
89 <div id="result"></div>
90
91 </div>
92
93 <script>
94
95 function checkScholarship(){
96
97   let marks=Number(document.getElementById("marks").value);
98
99   let scholarship="";
100
101   if(marks<0 || marks>100){
102     scholarship="Please enter valid marks.";
103   }
104   else if(marks<50){
105     scholarship="Not Eligible";
106   }
107   else if(marks<=69){
108     scholarship="Bronze Scholarship";
109   }
110   else if(marks<=84){
111     scholarship="Silver Scholarship";
112   }
113   else if(marks<=94){
114     scholarship="Gold Scholarship";
115   }
116   else{
117     scholarship="Platinum Scholarship";
118   }
119
120   document.getElementById("result").innerHTML=
121     "<b>Result:</b><br>"+scholarship;
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Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Output:

Screenshot of a web browser showing a Scholarship Eligibility form. The form is titled "Scholarship Eligibility" and asks the user to "Enter Marks (0-100)". The user has entered "89". A blue button labeled "Check Eligibility" is visible. Below the button, the result is displayed: "Result: Gold Scholarship".

Result/Conclusion: The Student Grading System successfully demonstrates the use of **HTML for structure**, **CSS for styling**, and **JavaScript for logic and validation**. It applies control structures, loops, arrays, and DOM manipulation to process user input and generate results. This project improves the accuracy of grade calculation, reduces manual errors, and provides immediate feedback, making it a simple and effective web-based application for evaluating student performance.